

Laptop Request Management System

Project Documentation

ServiceNow Application

1. Introduction

Project Title: Laptop Request Management System

Team Member: Guraka Murali

The Laptop Request Management System is a ServiceNow-based application designed to digitize and automate the process of requesting, approving, processing, and tracking laptop requirements for employees.

2. Project Overview

Purpose: To provide a centralized and transparent process for employees to submit laptop requests and for managers and IT teams to review, approve, process, and track those requests.

Key Features:

- Laptop request form for employee requests.
- Requester, employee email, laptop type, reason, manager, date, and status information.
- Manager approval and rejection workflow.
- Automated workflow using ServiceNow Flow Designer.
- Business Rule to prevent duplicate laptop requests for the same user within the current calendar year.
- Role-based access using IT Employee and IT Manager roles.
- ACL-based control for create, read, and write access.
- Approval records and request status tracking.
- Email/flow notifications related to approval processing.

3. Architecture

The application is implemented within the ServiceNow platform. Employees interact with the request form/catalog interface, while ServiceNow tables store request information. Business Rules perform validation, Flow Designer manages approval and workflow actions, and ACLs and roles provide access control.

User Interface: ServiceNow Laptop Request Form and Employee Service Catalog/Record Producer.

Application Layer: ServiceNow application logic, Business Rules, Flow Designer, and approval processing.

Data Layer: Laptop Request Form table containing request, employee, laptop, status, and manager information.

Security Layer: IT Employee and IT Manager roles with ACLs controlling access to records and operations

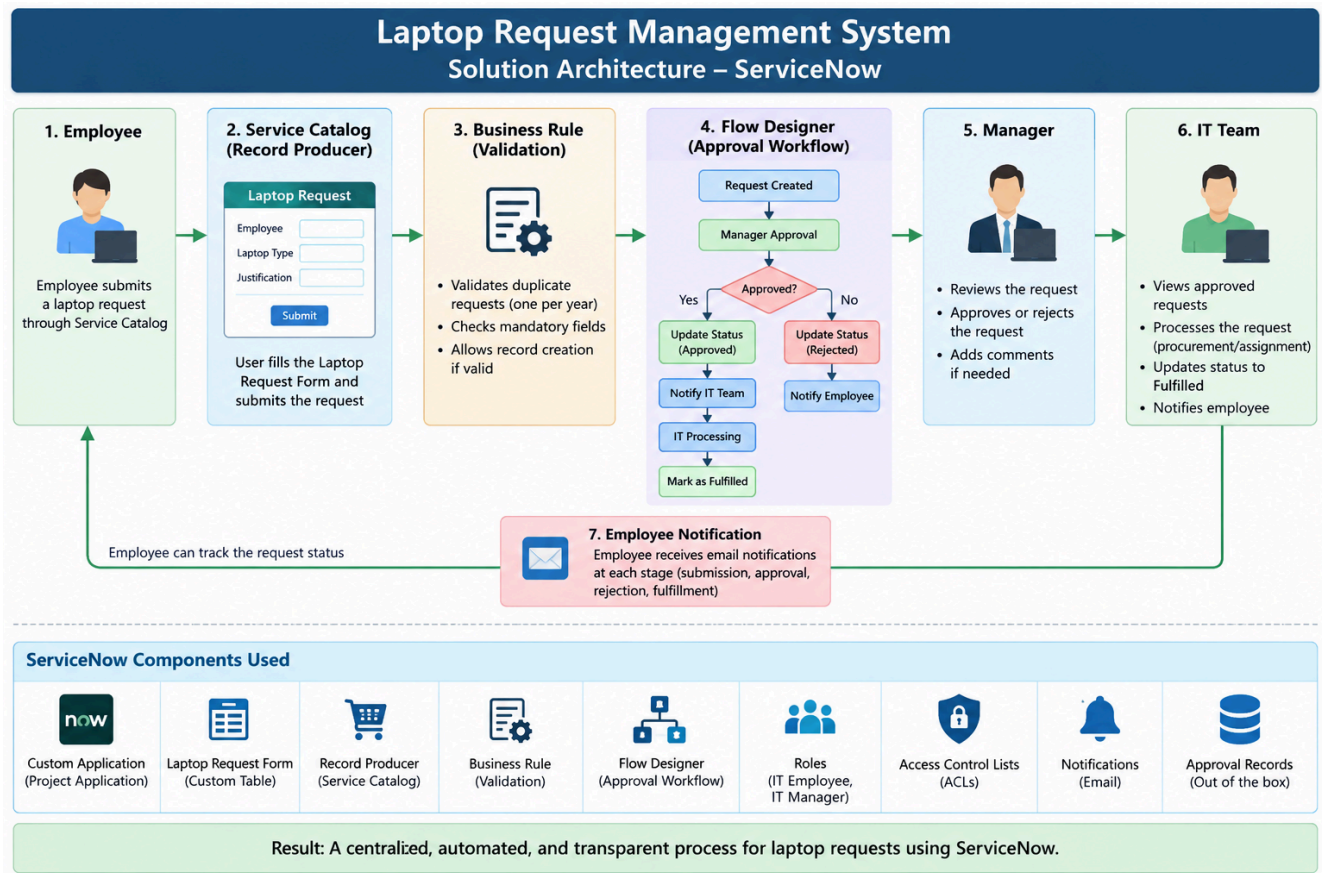


Figure 1: Solution Architecture Diagram – Laptop Request Management System

4. Setup Instructions

The application is configured and executed in a ServiceNow instance. The following requirements are needed to access and use the project:

- A valid ServiceNow instance and user account.
- Appropriate ServiceNow roles/permissions for the required activities.
- Access to the custom Laptop Request Management application.
- Access to the Laptop Request Form, Record Producer, workflow, and related records.

Security Note: User credentials and passwords are not included in this document.

5. Folder Structure

The project documentation is organized by project phase. The documentation package contains project planning, design, development/testing, and final documentation material.

- Project Planning – backlog, tracker, velocity, planning logic, and burndown chart.
- Project Design – solution architecture, proposed solution, and problem–solution fit.

- Project Development – User Acceptance Testing and Performance Testing.
- Project Documentation – FSD documentation and final report.

Project documentation folder structure:

Project Documentation/
 └─ FSD Documentation
 └─ Final Report

Project Development Phase/
 └─ User Acceptance Testing/
 └─ Performance Testing/

6. Running the Application

The application is accessed through the organization's ServiceNow instance. After logging in with an authorized account, the user can open the Laptop Request Management application and submit or process requests according to their assigned role.

1. Log in to the authorized ServiceNow instance.
2. Open the Laptop Request Management application.
3. For an employee request, open the Laptop Request Form/Record Producer.
4. Enter the required employee and laptop information and submit the request.
5. The configured validation checks the request and prevents duplicate requests when applicable.
6. The approval workflow routes the request for manager approval.
7. The manager approves or rejects the request.
8. IT users process approved requests and update the request status as required.
9. Users can review request and approval information from the relevant ServiceNow records.

The screenshot displays the ServiceNow interface for the 'Laptop Request Form - LAP001006'. The form contains the following fields:

- Number: LAP001006
- Status: Pending Approval
- Short description: [Empty]
- Date: 2026-08-17
- Employee Email: itil@example.com
- Requester: Abraham Lincoln
- Type of Laptop: Laptop
- Reason: -- None --
- Manager: [Empty]

Below the form is an 'Approvals' table with the following data:

State	Approver	Comments	Approval for	Created
Requested	Abel Tuter		(empty)	2026-08-17 11:39:07
Requested	Abel Tuter		(empty)	2026-08-17 11:39:04

Figure 2: Laptop Request Submission Screen

7. API Documentation

Not Applicable. The current version of the Laptop Request Management System does not implement a separate custom REST API or external backend API. The application uses ServiceNow's platform capabilities, tables, Business Rules, Flow Designer, approvals, roles, and ACLs.

8. Authentication

Access to the application is controlled through the ServiceNow platform's user accounts and role-based authorization. Different project roles are used to control the operations available to employees and IT managers.

IT Employee Role: Provides authorized IT employee access to relevant laptop request operations.

IT Manager Role: Provides manager-level access required for reviewing and approving requests.

ACLs: Create, read, and write access is controlled using configured Access Control Lists.

The screenshot shows the ServiceNow interface for configuring a role. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main header indicates the role being edited: 'Role - x_1959161_projec_0.it_employee'. Below this, the role configuration form is displayed with fields for 'Suffix' (set to 'it_employee'), 'Name' (set to 'x_1959161_projec_0.it_employee'), 'Application' (set to 'Project Application'), 'Elevated privilege' (unchecked), 'Assignable by' (empty), and 'Description' (set to 'Create and manage laptop requests'). Below the form are 'Update' and 'Delete' buttons. The 'Contains Roles' tab is selected, showing a search bar and a table with the role 'x_1959161_projec_0.it_employee'. The table is empty, displaying a 'No records to display' message with a question mark icon.

Figure 3: IT Employee Role Configuration

servicenow All Favorites History Workspaces **Role - x_1959161_projec_0.it_manager** Search

Role **x_1959161_projec_0.it_manager** Update Delete

* Suffix Application **Project Application**

Name **x_1959161_projec_0.it_manager** Elevated privilege ☐

Assignable by

Description **View-only access for managers**

Update Delete

Contains Roles Applications with Role Modules with Role Custom Tables

for text Search New Edit...

Role = x_1959161_projec_0.it_manager

Contains

No records to display

Figure 4: IT Manager Role Configuration

servicenow All Favorites History Workspaces **Access Control - x_1959161_projec_0.number** Search

Access Control **x_1959161_projec_0.number**

Type **record** Application **Project Application**

Operation **create** Active ☒

Admin overrides ☒ Advanced ☐

Decision Type **Allow If**

Protection policy **-- None --**

Name **x_1959161_projec_0.number**

Description **Allow create for records in x_1959161_projec_0.number, for users with role x_1959161_projec_0.it_employee.**

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role 1 to 1 of 1

Role
x_1959161_projec_0.it_employee

Security Attribute Condition

Local or Existing Existing

Figure 5: ACL Configuration for Create Access

servicenow All Favorites History Workspaces : Access Control - x_1959161_projec_0_number Search

Access Control
x_1959161_projec_0_number

Type: record
Operation: read
Application: Project Application
Active: ☒
Admin overrides: ☒
Decision Type: Allow If
Protection policy: -- None --
Name: x_1959161_projec_0_number
Description: Allow read for records in x_1959161_projec_0_number, for users with role x_1959161_projec_0.it_manager.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role: 1 to 1 of 1

Role
x_1959161_projec_0.it_manager

Security Attribute Condition
Local or Existing: Existing

Figure 6: ACL Configuration for Read Access

9. User Interface

The user interface is provided through ServiceNow forms, lists, and catalog/request components. The main Laptop Request Form captures the information needed to process a request.

- Laptop Request Form
- Employee Service Catalog / Record Producer
- Laptop request list and individual request records
- Approval related list
- Manager approval interface
- Request status and tracking information

Build and modify items faster with the improved [Catalog Builder](#).

Name: Laptop Request

Application: Project Application

* Table name: Laptop Request Form [x_1959161_projec,...]

Active: ☒

State: -- None --

Checked out: -- None --

Owner: System Administrator

What it will contain | Accessibility | Generated Record Data | Portal Settings

Add a short description and a full description

Short description:

Description:

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Applicable only for keyword search when AI Search is enabled.

Figure 9: Laptop Request Record Producer

Laptop Request Form LAP001006

Number: LAP001006

Status: Approved

Short description:

Date: 2026-08-17

Employee Email: itil@example.com

Requester: Abraham Lincoln

Type of Laptop: Laptop

Reason: -- None --

Manager:

Update Delete

Laptop request State Search

Approvals

State	Approver	Comments	Approval for	Created
Approved	Abel Tuter		(empty)	2026-08-17 11:39:07
Requested	Abel Tuter		(empty)	2026-08-17 11:39:04

1 to 2 of 2

Figure 10: Laptop Request Record with Approval Records

10. Testing

Testing was performed to verify the main functional workflow and project behavior. The project documentation includes User Acceptance Testing and System Performance Testing.

User Acceptance Testing: Covers request submission, manager approval/rejection, notifications, IT processing, laptop assignment, request tracking, and duplicate request validation.

Performance Testing: Reviews request data, data validation, record filtering, and the availability of dashboard/report functionality.

Security Testing: Role and ACL configuration was reviewed for controlled access to request operations.

UAT Result Summary: 8 project-specific UAT test cases are documented in the UAT report. The final Pass/Fail/Not Tested values should reflect the actual executed tests.

11. Screenshots or Demo

Screenshots and a project demonstration can be used to showcase the main ServiceNow features and workflow. Recommended screenshots include:

- Project application overview.
- Laptop Request Form.
- Record Producer – Laptop Request.
- Business Rule for duplicate request validation.
- Duplicate request validation result.
- Flow Designer approval workflow.
- Approval record / manager approval.
- IT Employee and IT Manager roles.
- ACL configuration.
- Laptop request record and approval related list.

The screenshot displays the 'Custom Application - Project Application' configuration page in ServiceNow. The interface includes a top navigation bar with 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main content area is divided into several sections: 'Application Scoping' with fields for 'Name' (Project Application) and 'Version' (1.0.0); 'Design and Runtime' with 'JavaScript Mode' (ECMAScript 2021 (ES12)) and 'Runtime Access Tracking' (Tracking); 'Subscription Management' with 'Licensable' (checked), 'Subscription requirement' (Monitor), and 'Subscription Model' (Not applicable); 'Primary menu' with a 'Menu' field; and 'End user access' with a 'User role' field. Each section contains descriptive text and a search icon.

Figure 11: ServiceNow Project Application

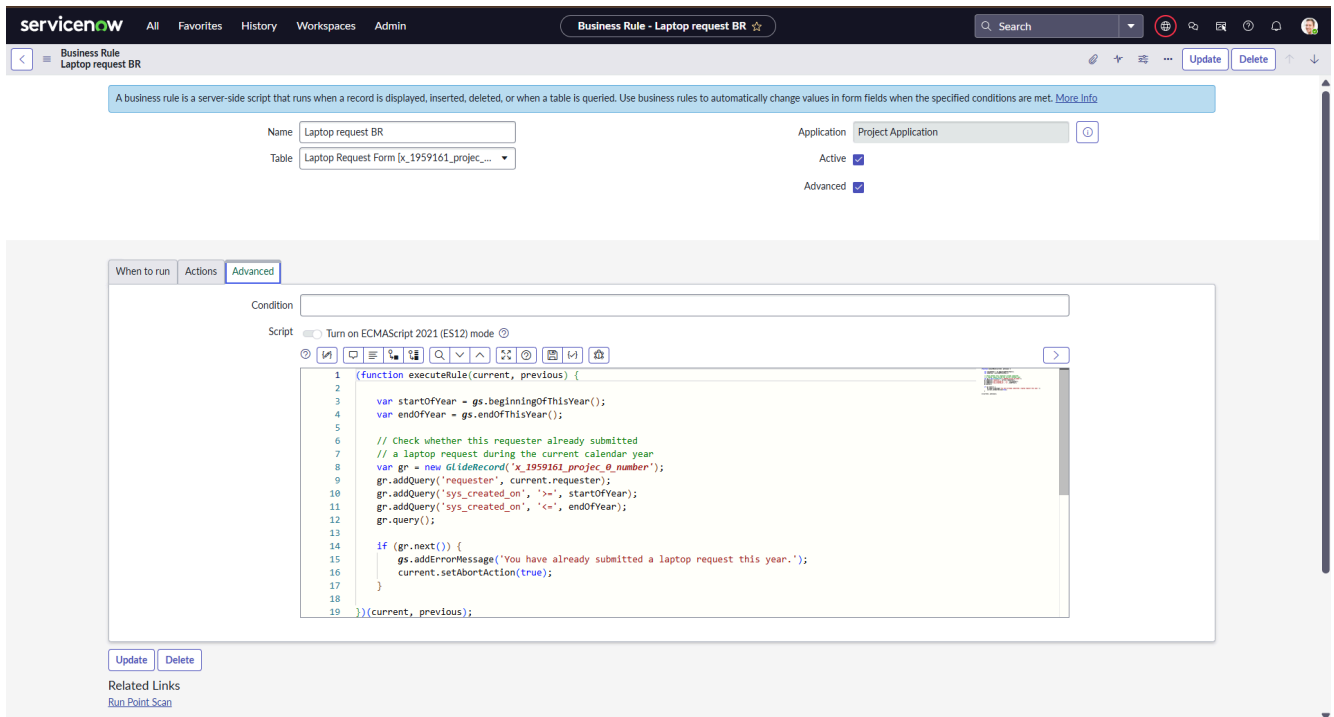


Figure 12: Business Rule for Duplicate Request Validation

Laptop Request Form - Create LAP001021

⊗ You have already submitted a laptop request this year. ×

⊗ Invalid insert

Number: LAP001021
Status: -- None --
Short description:
Date:
Employee Email:
Requester: Abraham Lincoln
Type of Laptop: -- None --
Reason: -- None --
Manager:
Submit

Figure 13: Duplicate Laptop Request Validation Result

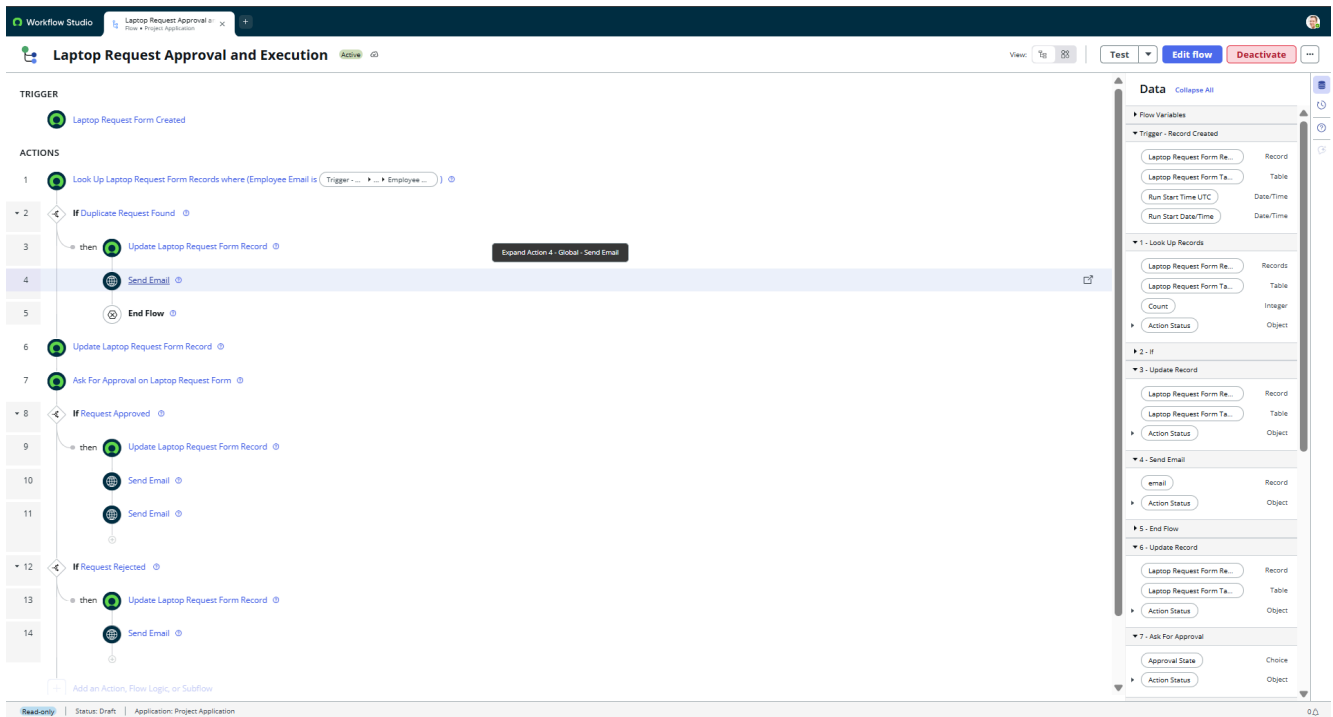


Figure 14: Laptop Request Approval and Execution Flow

The screenshot shows the "Relationship - Laptop request" configuration page in ServiceNow. The configuration details are as follows:

- Name**: Laptop request
- Application**: Project Application
- Applies to table**: Laptop Request Form [x_1959161_proj...]
- Queries from table**: Approval [sysapproval_approver]

Below the configuration fields, there is a script editor with the following code:

```

1 current.addQuery('source_table', parent.getTableName());
2 current.addQuery('document_id', parent.sys_id);

```

At the bottom of the page, there is a section for "Related Links" with a link to "Run Point Scan". Below this is a table for "External Related Lists" with the following structure:

Relationship = Laptop request	Related List	Application
No records to display		

Figure 15: Approval Relationship Configuration

Approval
Laptop Request Form: LAP001006

Approver: Abel Tuter

State: Requested

Approval Reason: Laptop request requires manager approval

Comments:

Post

Activities: 1

System Administrator

Field changes • 2026-08-17 11:39:07

Approver: Abel Tuter
State: Requested

Update Approve Reject Delete

Summary of Item being approved

Laptop Request Form

Number: LAP001006

Status: Pending Approval

Short description:

Date: 2026-08-17

Employee Email: itil@example.com

Requester: Abraham Lincoln

Type of Laptop: Laptop

Reason:

Figure 16: Manager Approval Request

servicenow All Favorites History Workspaces Admin

Laptop Request Form - LAP001006

Application scope: Project Application
Update set: Default (Project Application)

Search

Laptop Request Form LAP001006

Number: LAP001006

Status: Approved

Short description:

Date: 2026-08-17

Employee Email: itil@example.com

Requester: Abraham Lincoln

Type of Laptop: Laptop

Reason: -- None --

Manager:

Update Delete

[SN Utils] Versions (0)

Laptop request State Search

Approvals

State	Approver	Comments	Approval for	Created
Approved	Abel Tuter		(empty)	2026-08-17 11:39:07
Requested	Abel Tuter		(empty)	2026-08-17 11:39:04

1 to 2 of 2

Figure 17: Laptop Request with Approved Status

Demo Video: Insert the final project demonstration video link here after uploading the video.

12. Known Issues

The current version does not include a separate dashboard or story/report visualization. It also does not implement a separate custom API layer. These are documented as Not Applicable in the Performance Testing document rather than being treated as defects.

Any additional issue discovered after final testing should be added here with its description, impact, and status.

13. Future Enhancements

- Add a dedicated dashboard showing request, approval, and fulfillment statistics.
- Add reports for pending, approved, rejected, and completed requests.
- Extend the system to support additional equipment such as monitors, keyboards, and mobile devices.
- Integrate laptop inventory/asset information for automatic availability checks.
- Add more advanced notifications and reminders for pending approvals.
- Add additional approval rules based on department, location, laptop type, or cost.
- Provide enhanced request analytics and management views.

Conclusion

The Laptop Request Management System provides a structured ServiceNow-based solution for managing employee laptop requests. It combines request submission, validation, manager approval, role-based security, workflow automation, notifications, and request tracking within a centralized platform. The documentation can be extended with final screenshots and the project demonstration link.

References

- ServiceNow platform documentation and product guidance.
- ServiceNow Flow Designer documentation.
- ServiceNow Access Control List (ACL) documentation.
- Project-specific User Acceptance Testing document.
- Project-specific Performance Testing document.